



CEO Turnover and Executive Pay Dispersion in Europe

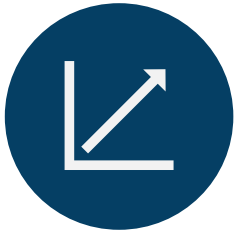
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- 1 Motivation, Research Question, and Economic Relevance**
- 2 Preparation of Final Analysis Panel and Pay Dispersion**
- 3 Data Visualizations**
- 4 Methodology**
- 5 Results**
- 6 Synthesis and Limitations**

Motivation, Research Question, and Economic Relevance



Motivation

A CEO turnover is a major corporate governance event that can reshape a firm's strategic direction, leadership priorities, and internal incentive structure. Because CEO succession often coincides with renewed compensation decisions, turnover may affect both the level of CEO pay and the relative pay gap between the CEO and other senior executives.



Research question

This study examines whether CEO turnover causally changes CEO compensation and CEO-to-executive pay dispersion within European firms.



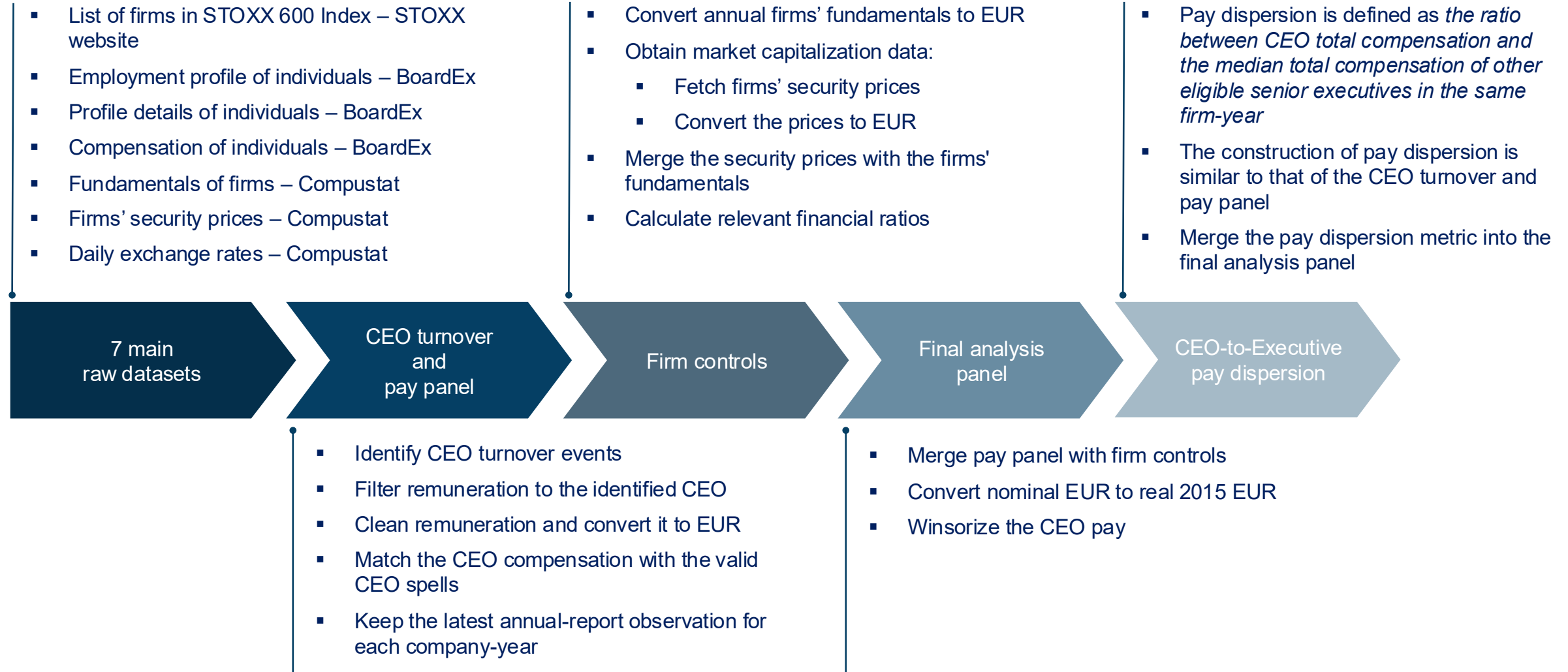
Economic Relevance

CEO turnover may reset compensation incentives and reshape pay inequality within top management. By examining both CEO pay and CEO-to-executive pay dispersion, the study shows whether leadership transitions affect not only how much CEOs are paid, but also how rewards are distributed among senior executives.



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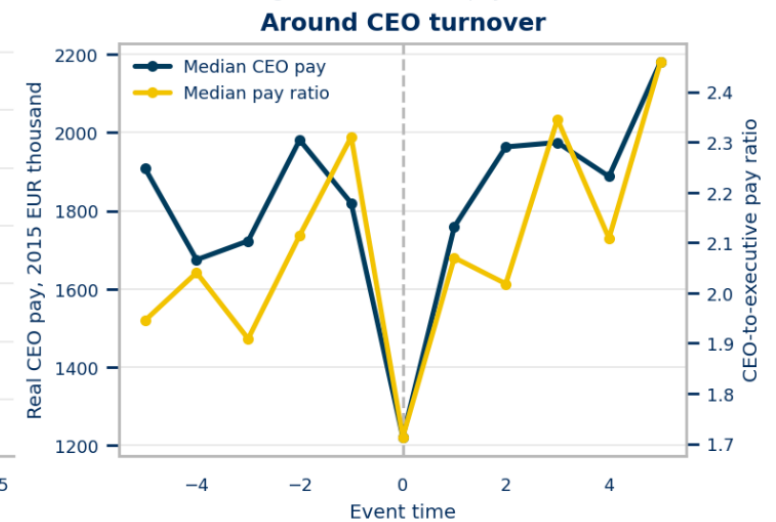
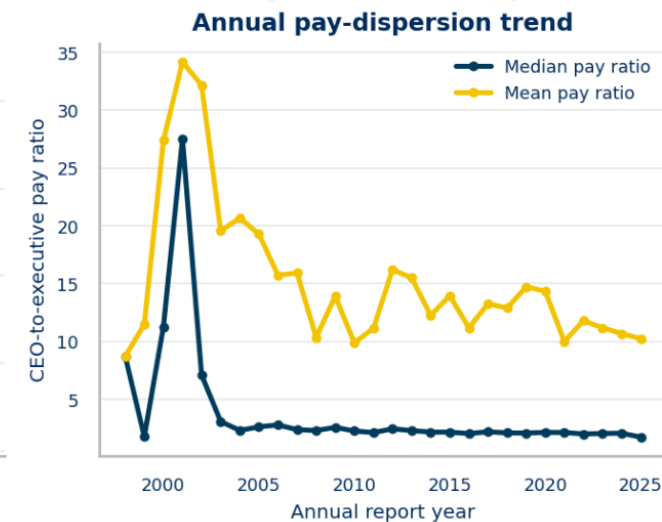
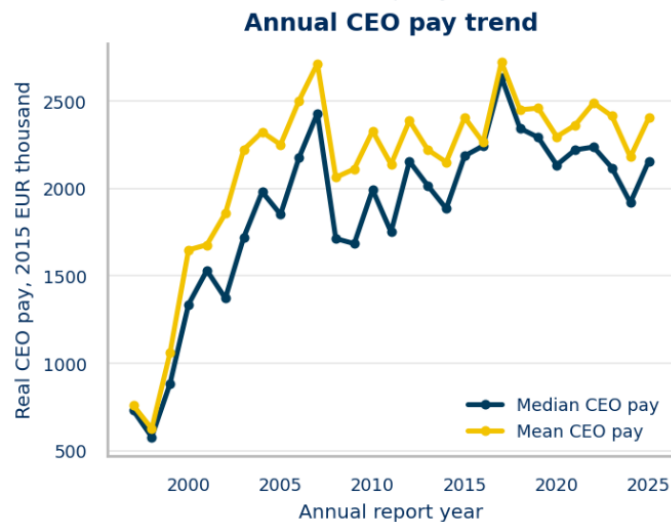
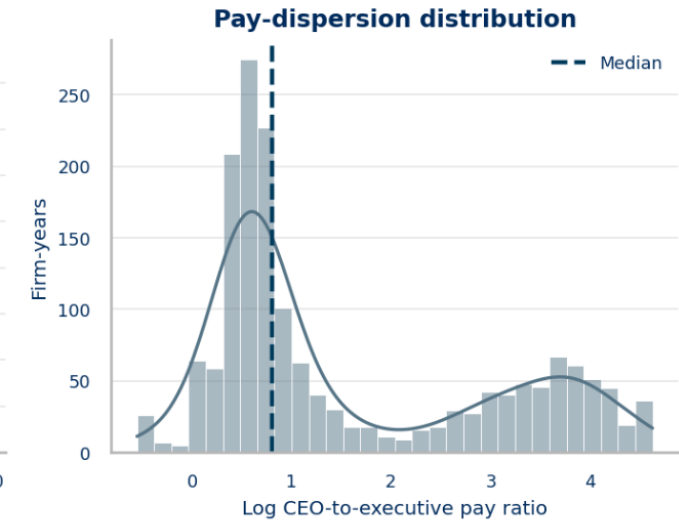
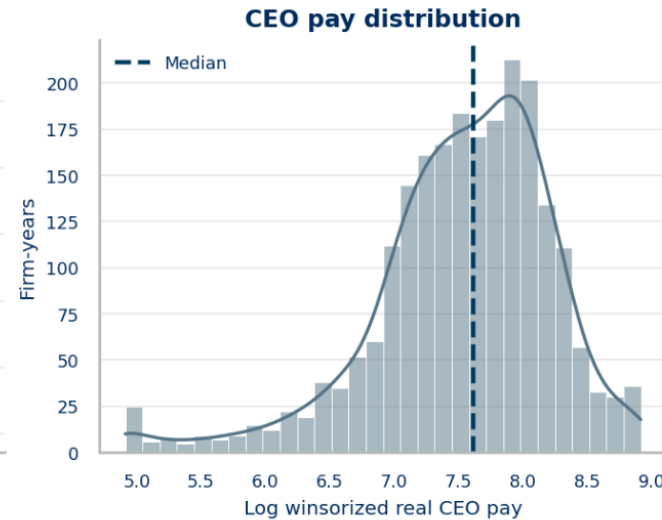
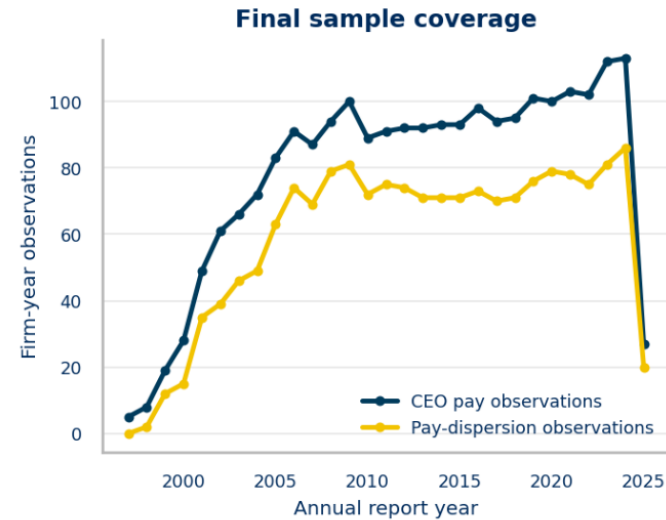
Preparation of Final Analysis Panel and Pay Dispersion





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Data Visualizations





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Causal Inference

- Goal: estimate effect of CEO turnover
- Outcomes:
 - log CEO pay
 - log CEO-to-executive pay dispersion
- Treatment:
 - CEO turnover
 - post-turnover period used for DiD timing
- Method:
 - Difference-in-Differences
 - event study around turnover year
- Controls:
 - log assets, ROA, leverage, country, sector, year fixed effects

Supervised Learning

- Goal:
 - predict pay, turnover, and dispersion
- Tasks:
 - CEO pay regression
 - turnover classification
 - pay-dispersion regression
- Design:
 - train: before 2020
 - test: 2020 onward
- Models:
 - Ridge for regression
 - Logistic Regression for turnover
 - Random Forest, XGBoost

Unsupervised Learning

- Goal: identify compensation profiles
- Inputs:
 - CEO pay, pay dispersion, firm fundamentals
 - CEO demographics
 - turnover indicators
 - country, sector, year
- Method:
 - preprocessing pipeline
 - PCA
 - K-Means clustering
- Validation:
 - silhouette score for k selection
 - cluster profile comparison
 - robustness without turnover indicators



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Causal Inference

- Main finding:
 - CEO turnover does not produce a persistent change in CEO pay or pay dispersion.
 - CEO pay drops temporarily in the turnover year.
- CEO pay:
 - no significant average post-turnover effect
 - DiD coefficient = -0.073, $p = 0.164$
 - turnover-year event effect: -0.598
- Pay dispersion:
 - no significant post-turnover effect
 - DiD coefficient = 0.005, $p = 0.958$
 - event-study effects are not statistically significant
- Interpretation:
 - turnover creates a short-run CEO pay dip, but no durable reset in compensation structure

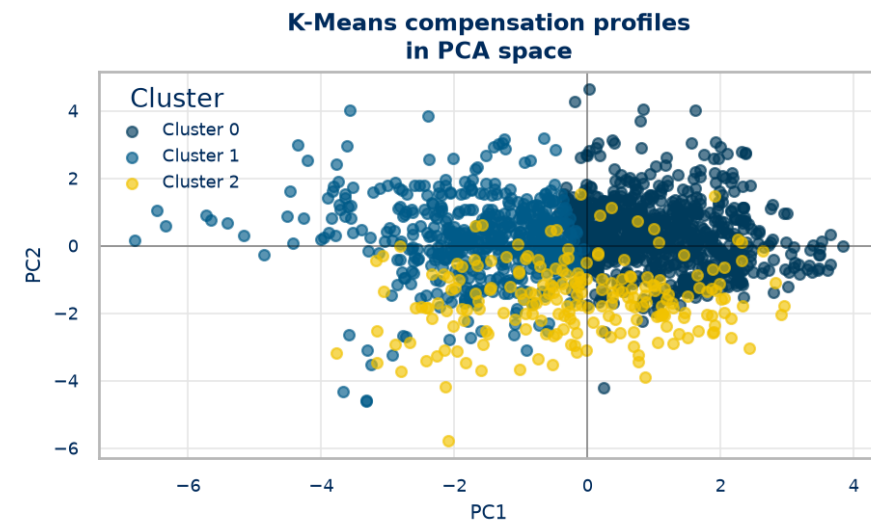
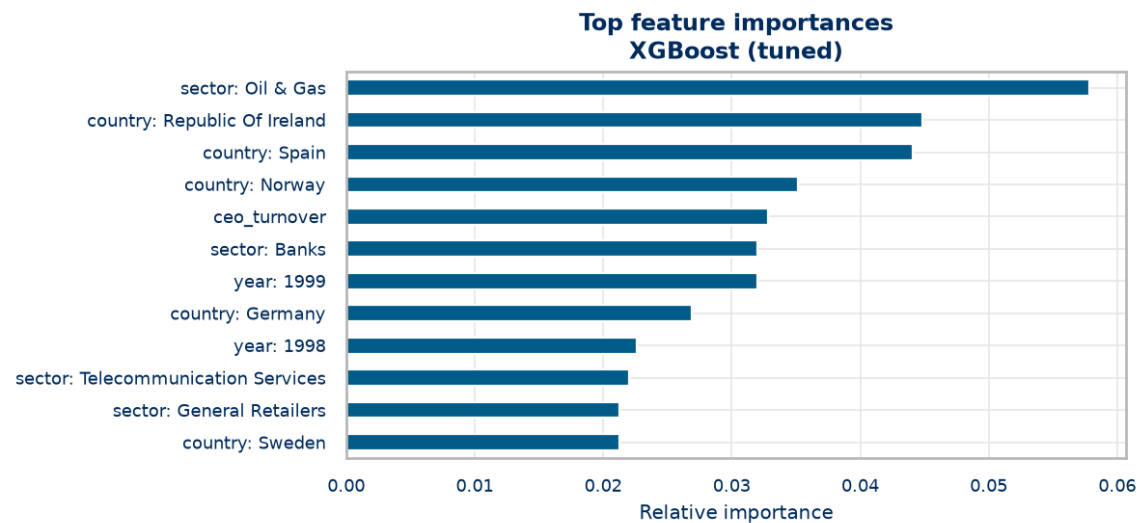
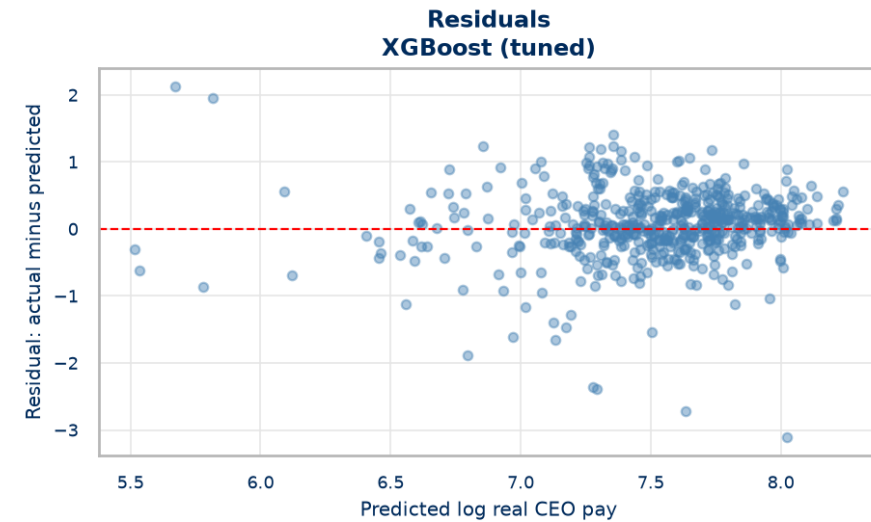
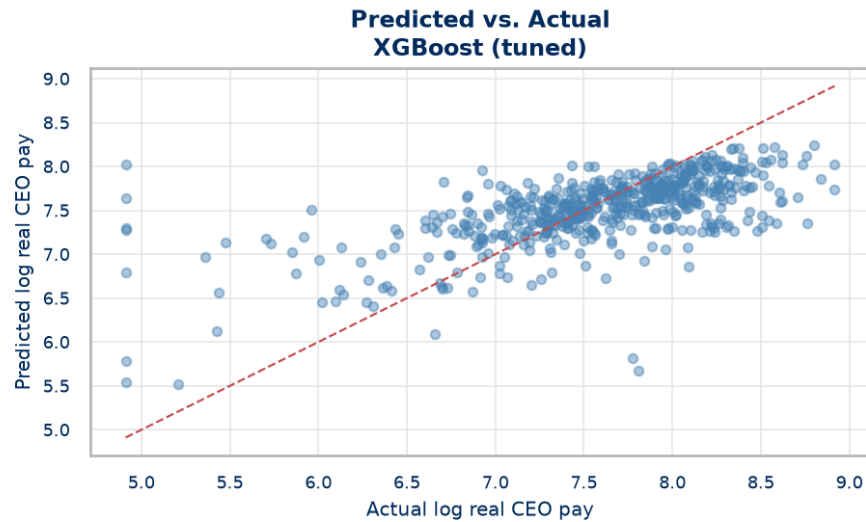
Supervised Learning

- Main finding:
 - CEO pay and pay dispersion are predictable to some extent
 - CEO turnover is difficult to predict
- CEO pay prediction:
 - best model: XGBoost
 - RMSE = 0.529
 - $R^2 = 0.392$
- Turnover classification:
 - best model: Logistic Regression
 - ROC-AUC = 0.597
 - MCC = 0.07
- Pay dispersion prediction:
 - best model: XGBoost
 - RMSE = 0.818
 - $R^2 = 0.634$

Unsupervised Learning

- Main finding:
 - clusters describe compensation profiles, but should not be interpreted causally
- Best solution:
 - $k = 3$
- Clusters broadly distinguish:
 - different compensation and firm-size profiles
 - turnover timing partly contributes to the separation
- Robustness:
 - removing turnover indicators changes the best solution to $k = 2$
 - cluster separation remains, but turnover-state alignment weakens
- Interpretation:
 - useful descriptive evidence, not independent causal evidence

Results





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- **Synthesis**

- CEO turnover does not create a persistent change in CEO pay or pay dispersion.
- The clearest effect is a temporary CEO-pay drop in the turnover year.
- Pay and pay dispersion are partly predictable, but turnover itself is hard to predict.
- Clusters reveal compensation profiles, but they are descriptive, not causal.

- **Limitations**

- Limited turnover sample reduces event-study power.
- Annual pay data cannot capture within-year contract changes.
- Pay-dispersion coverage depends on available CEO and non-CEO executive remuneration.
- Unobserved governance factors may affect turnover timing.
- European disclosure differences may create measurement error.
- ML and clustering results are predictive/descriptive, not causal.